A Quantitative Analysis of Cultural Universals and Variation in Moral Association

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Introduction

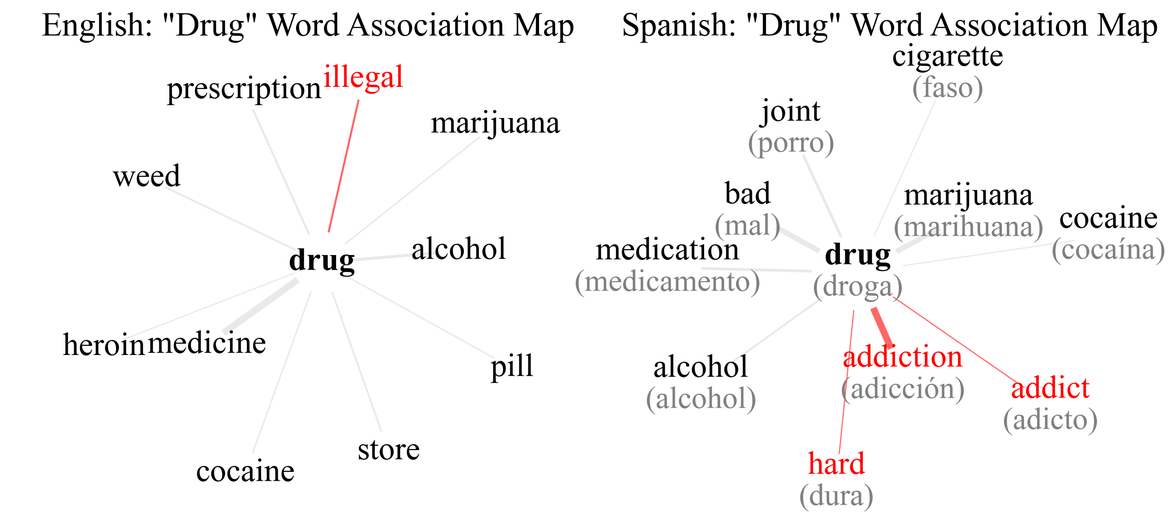


Figure 1: English and Spanish associations for the word drug (droga in Spanish) (De Deyne et al., 2019, Cabana et al., 2024). Red responses are words in the Moral Foundations Dictionary (MFD) (Graham et al., 2009; Frimer et al., 2019).

- Cultural variation in moral perception can affect moral decision-making (Awad et al., 2018)
- Recent work shows word associations can effectively capture English speakers' moral perception of certain concepts
- **We present a framework beyond existing work to characterize moral associations across cultures through word associations collected from speakers of different languages**. We propose that examining cross-linguistic patterns of word association may help uncover both shared and varied moral views among speakers.

Data

(Collected in English and Spanish)

- **Word Association Data** (De Deyne et al., 2019, Cabana et al., 2024)
- **Moral Foundations Dictionary 2.0** (Graham et al., 2009; Frimer et al., 2019)
  - Translated the English MFD to Spanish
- **Wordnet** (Bond & Foster, 2013)

Conclusion

Our analysis of English & Spanish word association data reveals shared and language-specific tendencies in moral association.

We found that certain semantic domains, such as 'wrongdoer' and 'belief', universally predict moral association, while other domains exhibit language-specific associations.

At the structural level, we observe that local network cohesion predicts moral relevance across both languages, while global centrality and node influence measures have differential effects.

Methods

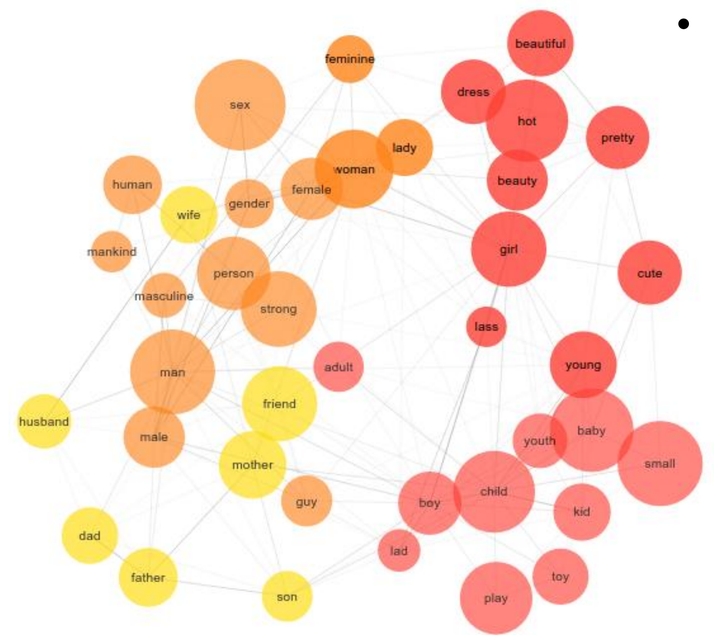


Figure 2: English Word Association Network created by linking cue and response words associations (De Deyne et al., 2019)

- Create additional features for word association data that examine the role of the word in the network
- Ascribe each cue word with a corresponding category from Wordnet dataset

  Ex. drug → agent → causal agent → physical entity → **entity**
- Calculate the MAG score to capture the proportion of morally-relevant associations for a given concept in the word association network (Ramezani & Xu, 2025)

$$\mathrm{MAG}_{(C)} = \frac{\sum_{t \in T(C) \cap M} A_{c,t}}{\sum_{t \in T(C)} A_{c,t}}$$

Results

We used ordinary least squares regression with LASSO cross-validation to examine which graph-based and semantic features predict moral association:

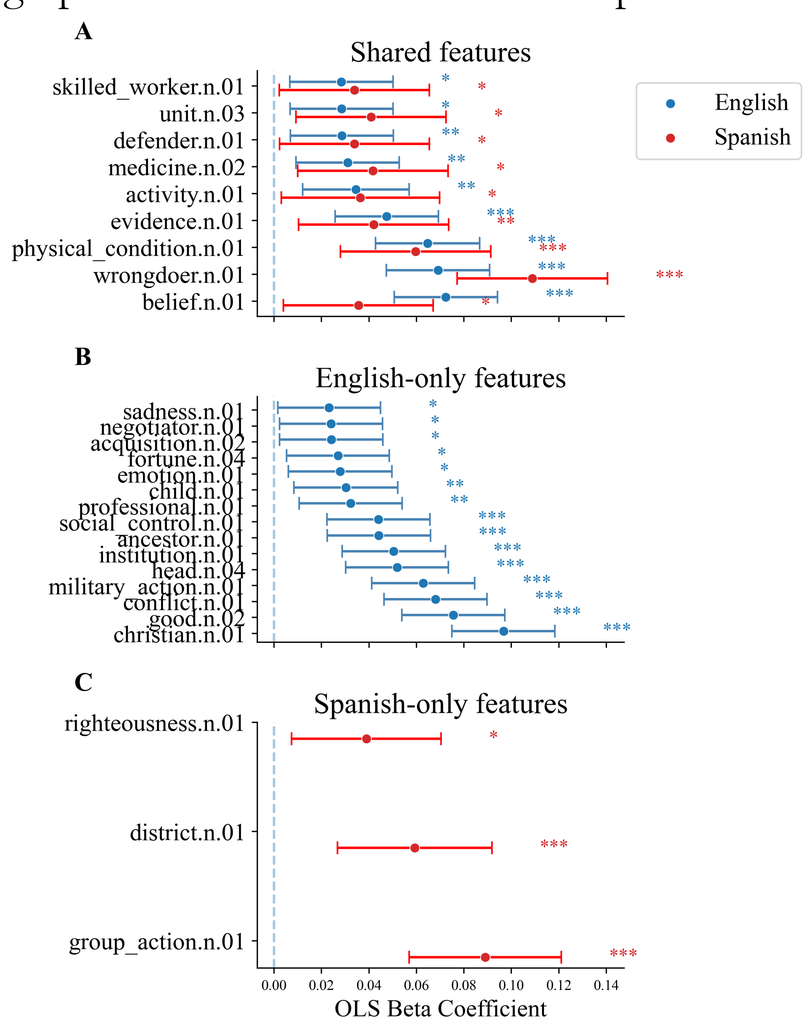


Figure 3: WordNet categories predicting MAG scores (all $p < 0.05$, positive β coefficients). (A) Shared categories between English and Spanish, ranked by β coefficient magnitude. (B, C) Language-specific categories. Asymmetry between English (B) and Spanish (C) categories might reflect differences in data coverage or lexical diversity.

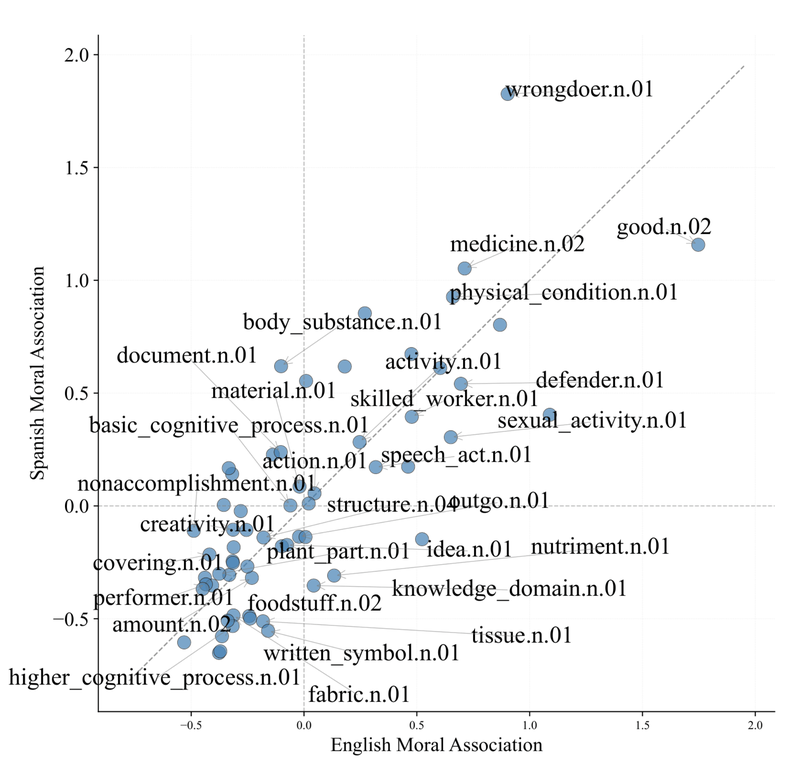


Figure 4: Moral association scores for WordNet categories. Each point represents a category containing at least 10 members in each language. The diagonal reference line show perfect cross-linguistic agreement. Points above the line show stronger moral associations in Spanish, and points below show stronger moral associations in English.

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